1 Two parallel cylindrical conductors each have a small cross-sectional area A. A thin metal bar connects the two conductors, as shown in Fig. 1.1.

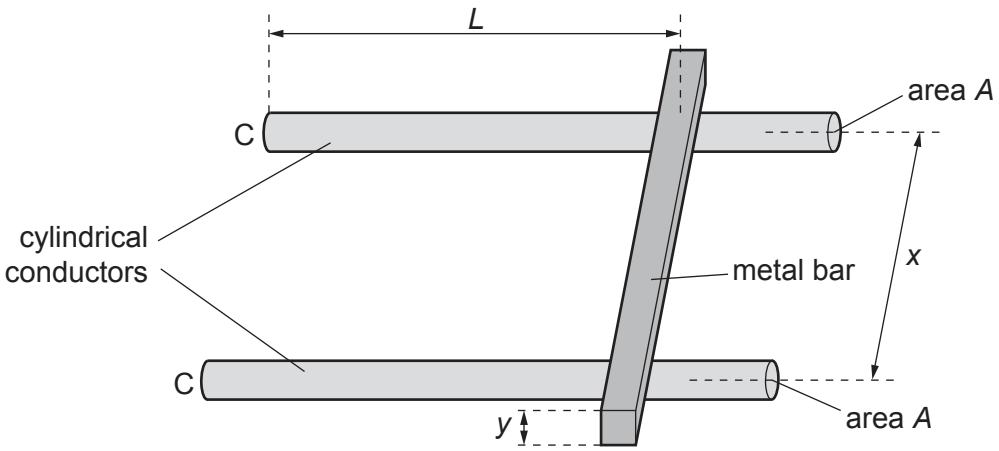

Fig. 1.1 (not to scale)

The metal bar has a square cross-section with sides of length y. For each conductor, the distance between its end C and the centre of the metal bar is L. The distance between the centres of the conductors is x.

The ends C are connected to a power supply and the current I in the conductors is measured.

It is suggested that I is related to L by the relationship

$$\frac{E}{I} = \frac{2PL}{A} + \frac{Qx}{y^2}$$

where E is the electromotive force (e.m.f.) of the power supply, and P and Q are constants.

Plan a laboratory experiment to test the relationship between I and L.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for P and Q.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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